

**ASE2910 Applied Linear Algebra / AUS2910 Fundamental Math for AI
Homework #2**

- 1) *Angle between two vectors.* Show that any two vectors $a, b \in \mathbb{R}^n$ satisfy the following, where θ is the angle between a and b . You may provide a proof for the two-dimension case, which easily generalizes to general n -dimension cases.

$$a^T b = \|a\| \|b\| \cos \theta.$$

- 2) *Orthogonal projection.* Let x be an n -vector and $u_1, \dots, u_k$ with $k < n$ be orthonormal n -vectors. The projection of x onto the span of $u_1, \dots, u_k$ is

$$\hat{x} = (u_1^T x)u_1 + \dots + (u_k^T x)u_k \in \mathbb{R}^n$$

and the projection of x onto the span of $u_1, \dots, u_{k-1}$ is

$$\tilde{x} = (u_1^T x)u_1 + \dots + (u_{k-1}^T x)u_{k-1} \in \mathbb{R}^n.$$

Show that $\hat{x}$ is closer to x than $\tilde{x}$ in that

$$\|\hat{x} - x\| \leq \|\tilde{x} - x\|.$$

- 3) *A special k -means instance.* In the k means algorithm, we maintain cluster representatives $(z_1, \dots, z_k)$, each an n -vector, and at each iteration find the cluster representative z_i nearest a point x . Suppose that $k = 2$ and that we have cluster representatives $z_1, z_2 \in \mathbb{R}^n$, where $\|z_1\| = \|z_2\|$.

- a) Show that a point x is assigned to cluster 1, that is, $\|x - z_1\| < \|x - z_2\|$, if and only if

$$(z_1 - z_2)^T x > 0.$$

- b) In the case $n = 2$, draw a picture of your result. Your picture should contain the points z_1 and z_2 , the vector $z_1 - z_2$, and a line such that any point x falling on one side of the line will be assigned to cluster 1, and any point x falling on the other side of the line will be assigned to cluster 2. Label the sides of the line.

- 4) *VMLS Exercises.*

- a) **3.12** *Nearest point to a line.*
- b) **3.14** *Nearest unit vector.*
- c) **5.1** *Linear independence of stacked vectors.*
- d) **5.4** *Norm of linear combination of orthonormal vectors.*
- e) **5.5** *Orthogonalizing vectors.*